Informatics Institute of Technology

Business School

Assignment Cover Sheet

DOC334 – Computer Programming

Course	:	Foundation Certificate Programme
Unit Code and Description	:	DOC334 – Computer Programming
Module Leader	:	Mr. Nishan Saliya Harankahawa
Assignment Number	:	1
Assignment Type	:	Individual
Issue Date	:	2023-07-10
Deadline	:	on or before 11:59 PM (2023-07-31)
Qualifying mark	:	40%

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Read The Below Instructions Carefully

The department is **NOT RESPONSIBLE** if an assignment is lost. To make up for this eventuality, you are advised to keep multiple copies of the assignment OR to ensure you have the means of re-creating it. If not, you'll be allocated **zero** marks!

1. Procedure for Handling Work:
    a. Follow any specific instructions given on the assignment specification.
2. Penalties for Late Hand In:
    a. If students submit coursework late but within 24 hours of the specified deadline, the work will be marked and will then have 10% of the overall available marks deducted, to a minimum of the pass mark (40%).
    b. If students submit coursework more than 24 hours after the specified deadline, the work is **not** marked and will be given a mark of **zero** for the work in question.
3. Exceptional Factors Affecting your Performance:
    a. Students should submit written evidence to the Registrar's Department with a copy to the Module Leader of exceptional circumstances, which they consider having caused them to submit assessments late and for which they do not wish to attract any penalty. These must be handed over to the Registrar within four working days of the deadline.
    b. Make use of proper Python 3.x coding and language constructs for a better program. You should follow good programming techniques when completing this coursework.

Plagiarism

The strength of the university depends on academic and personal integrity. In this course, you must be honest and truthful. Plagiarism is the use of copied codes, text, someone else's work, words, or ideas as if they were your own.

PLAGIARISM IS A **SERIOUS OFFENSE** AND WILL NOT BE TREATED LIGHTLY.

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Deliverables

The following should be submitted.

1. A report including a description of the problem statement and the solution you have developed.
    a. Make sure you have followed the good practices of report writing covered by DOC311 module.
    b. Use good word processing skills learned from DOC314 module.
    c. Screenshots of the program output in various states must be included.
    d. All program codes must be included in your report as text (**NOT image screen captures**)
2. Test cases used to test the programs and the results must be included in your submission.
3. You need to provide the PDF version of your report and a ZIPPED folder which contains the full and working Python 3.x program.
    a. You can put both the program and the report to a single Zipped folder with your IIT student ID.
    b. For example, a zipped folder called **20229999** that carries PDF report and the final full Python program.
4. All codes must be written in Python 3.x version.
5. The completed coursework must be submitted **via the LMS assignment upload link**. **DO NOT** send it to your lecturer!
    a. **You can do only one upload**. If there are any issues with the upload, notify your lecturer before the deadline.

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Scenario

The “The Kingdom of Miranda” is a circular shaped small island with 5 main cities named as Alvin (Capitol), Jamz, Razi, Mali and Zuhar. The country looks like below.



Figure 1 : Kingdom of Miranda

The official currency of the country is KMD (Kingdom of Miranda Dollars). The country uses a single cab service known as DropMe™. The DropMe™ price chart for travelling is shown below. All the prices are in KMD.

City Name	***Alvin***	***Jamz***	***Razi***	***Mali***	***Zuhar***
Alvin	0	20	40	40	20
Jamz	20	0	20	40	40
Razi	40	20	0	20	40
Mali	40	40	20	0	20
Zuhar	20	40	40	20	0

Figure 2 : Trishwa Price Plan under DropMe™

Based on the above chart, price to travel from Alvin to Jamz is just 20 KMD while Razi to Razi is 0 KMD; which means the ride is free with in the same city.

The country uses only Trishaws, Cars and Vans for cab services. The default transport is Trishaws while passengers can ask for cars or vans if they prefer. Car prices have doubled while van prices have tripled.

For example, a trishaw ride from Jamz to Mali is 40 KMD while the same ride in a car is 80 KMD. It will be 120 KMD if you take a van.

The vehicles will always take the shortest path for traveling.

The president of The Kingdom of Miranda hired a good Python coder from IIT (You!) to develop a console application which will allow users to view trip details and generate invoices. The application is used by the passenger. The following commands can be issued to run the application from the client side.

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Command Types

Command	Description
C:\Users\Nishan>dm alvin razi	Shows the price between the two cities and generates an invoice file for the trip
C:\Users\Nishan>dm alvin razi /pro2	Shows the price between the two cities after applying the promo code. The promo code amount is deducted from the final bill value. This will generate an invoice file for the trip with promo details. 2 KMD is reduced here.
C:\Users\Nishan>dm /price	Show the full price plan for the whole country (for all 3 vhicles)
C:\Users\Nishan>dm alvin razi /c	Shows the price between the two cities and generates an invoice file for the trip. This uses a car (/c for car while /v for a van. Default is Trishaw)
C:\Users\Nishan>dm alvin razi /pro10 /v	Shows the price between the two cities while apply a 10 KMD reduction to total bill. The rider prefers a van.
C:\Users\Nishan>dm /?	Show how this “dm” command functions

The promo codes are available under especial circumstances or issued individually. The format of these codes is as below.

- pro2 (2 KMDs are reduced from the total bill)
- pro5 (5 KMDs are reduced from the total bill)
- pro10 (10 KMDs are reduced from the total bill)

The above three are samples while the promotion amounts can range from 1 KMD to maximum of 15 KMD.

The application itself will generate price reductions for customers who use the application. These auto-generated promotions are always at a fixed price of 5 KMD. These are **not taking place** if the passenger uses a promotional code. This feature is a random feature! Only the lucky passengers will get this special promotion time-to-time.

Every time a passenger uses the app, the invoice shown will be saved in a text file in the format. Three such sample files are shown below.

Date : 2023-06-26 Time: 06:28:29 Start : Alvin End : Razi Amount : 40 KMD Promo : 0 Random Reduction : 0 Final payment : 40 KMD	Date : 2023-06-26 Time: 06:30:29 Start : Alvin End : Razi Amount : 40 KMD Promo : 10 KMD Random Reduction : 0 Final payment : 30 KMD	Date : 2023-06-27 Time: 05:28:00 Start : Alvin End : Razi Amount : 40 KMD Promo : 0 Random Reduction : 5 KMD Final payment : 35 KMD

A file is saved for each invoice and the file name should look like 2023-06-26 06_30_29_4526.txt

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Special Notes

- This application will be a console application (**not a GUI** application). If a GUI application is built, **marks will be zero for the full ICW**.
- DO NOT use time.sleep() function in your program. This will reduce the marks for this component.
- The folder name must be IIT ID. If you put any other details to the folder name, marks will be reduced. For example,
    - 20229999 ✓
    - IIT-20229999 ✗
    - 20229999 jude ✗
    - 20229999 IIT ICW ✗

End of Coursework

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